

THE ENGINE ROOM

PART 1

MARINE ENGINEER

Anybody who wants to be a Marine Engineer must be ready and prepared to spend a long period in training. Practical training is of crucial importance because the sea environment is very demanding. Sea training is not easy, although it only takes place when a candidate has sufficient knowledge already. A cadet engineer has to learn new skills and put them into practice. For instance, he learns how to do machine repairs like opening up different bearings, tube sleeves, and rusty items like nuts and bolts, as well as cleaning valves and changing filters.

The successful candidate may be awarded a diploma and will then be able to work on ships as a qualified Marine Engineer class four. Then after some time spent working on ships, he can sit for competency certificates as a class-two Marine Engineer. Again, after some time at sea, he can sit for the class-one certificate of competency, which qualifies him to take up the job of the Chief Engineer on board a ship.

A marine engineer can be called a ship mechanic, a ship machinist, a ship engine operator, or a ship engine room attendant.

THE ENGINE DEPARTMENT

The Chief Engineer, or Chief Engineering Officer, is in charge of the Engine Department. He is responsible for all technical operations of the vessel, including engineering, electrical and mechanical units. In particular, he is responsible for all the propulsion machinery, power generating equipment and auxiliaries. He has to keep documents on the machinery working as well as all the repairs carried out on the vessel. He also logs fuel oil consumption. A varying number of officers, petty officers and ratings assist the chief engineer. The engine officers' hierarchy goes as follows:

- 1) The First Engineer. He is responsible for maintenance and operations of the engineering and technical units.

- 2) The Second Engineer. His responsibilities usually include the maintenance of lubricating systems, engine room auxiliaries, and electrical equipment.
- 3) The Third Engineer. He is usually responsible for fuel and water systems. He also supervises tanks soundings and monitors the boiler room equipment.
- 4) The Fourth Engineer. His responsibilities may include, for instance, the operation and maintenance of engine room auxiliaries.
- 5) The Motorman - His duties are defined by the head of the engine department and can include, for example, the daily maintenance and cleaning of specific engine parts.

The propulsion plant department can also include some petty officers, such as the donkey man and the storekeeper and, if the ship is a tanker, there may also be the pump man. The first one mentioned attends a donkey, auxiliary boiler, especially when the ship is in port. A storekeeper is in charge of all the spare parts and equipment stored for the engine room. The last one, a pump man, is employed to maintain and operate cargo pumps.

The engine room ratings, e.g. fire-fighters, greasers are usually employed on watches to assist the engineer in charge. They are responsible for daily cleanliness of the engine room and for routine oiling, greasing and machinery servicing.

WATCH KEEPING

The machinery driving a vessel which is underway is usually operated 24 hours a day. All running machinery must be controlled continuously in order to prevent any failure of the equipment. The majority of control systems on modern ships are automatic. A ship may operate for agreed periods with unmanned machinery, called UMS, which stands for Unattended Machinery Spaces.

The standard system of watches adopted on board is usually a four-hour period on duty followed by eight-hour rest. The word "watch" means both the period and the crew working at that time. The three watches in any 12 hour period are usually: 12 to 4, 4 to 8, 8 to 12. Thus, for instance, an engineer on duty for the 8 to 12 watch works from 8 a.m. to 12 noon and from 8 p.m. to 12 midnight.

A watch is usually made up of an engineer in charge with an assistant engineer and a rating. Their duties include inspecting the main propulsion plant, auxiliary machinery

and steering gear spaces. They should note any malfunctions and breakdowns, report and correct them.

Time at sea is expressed using the 24-hour system, as opposed to the 12-hour system commonly used on land.

Here are a few examples that illustrate the rule:

ON LAND	AT SEA
6.00 a.m. - six o'clock	0600 - six hours
1.30 p.m. - one thirty	1330 - thirteen hours thirty
5.25 p.m. - five twenty five	1725 - seventeen hours twenty five
midnight	0000 or 2400 - twenty four hours
noon or midday	1200 - twelve hours

Internal Combustion Engines

Internal-combustion engines are machines that convert heat into mechanical energy. In internal-combustion engines, burning of the fuel inside a tightly closed cylinder results in expansion of gases. The pressure created on top of a piston makes it move. The back-and-forth motion of a piston is known as the reciprocating motion (straight-line motion). This motion must be changed to rotating (turning) motion to perform a useful function, such as propelling a ship or driving a generator to produce electricity.

All internal-combustion engines rely on three things: fuel, air and ignition. Fuel contains energy for engine operation, air contains oxygen necessary for combustion, and ignition starts the process of combustion.

All internal-combustion engines consist of one or more cylinders that are closed off at one end and have a piston driving up the other end. Cylinders may be arranged either in a straight line (in-line) or in a V shape. When a piston slides downward as a consequence of the pressure of expanding gases inside a cylinder, the upper end of the connecting rod moves downward together with the piston. The lower end of the connecting rod moves down in a circular motion. This makes the crankshaft rotate.

There are different kinds of internal-combustion engines. The most commonly used nowadays are diesel and petrol engines. Diesel engines are extensively used in ship propulsion, and petrol engines in the automotive industry.

FOUR-STROKE CYCLE

The four-stroke engine was first introduced by Nikolaus Otto at the end of 19th century and since then it has also been known as the Otto cycle. The commonly used term, however, is “four-stroke”. It takes its name from the four strokes of the piston needed to complete the processes of converting fuel energy into work. The four strokes of the piston are known as the suction (intake or induction) stroke, the compression stroke, the power stroke and the exhaust stroke.

SUCTION . During this stroke, the crankshaft rotates clockwise and the piston moves down the cylinder. The inlet valve is open and a fresh air charge is drawn into the cylinder.

COMPRESSION. The inlet valve closes and the air charge is compressed by the piston moving up. Its pressure and temperature increase. By the time the piston approaches the cylinder top, known as Top Dead Centre (TDC), the pressure is over 100 bar.

POWER. Just before TDC, fuel is injected into the cylinder by the fuel injector. The fuel is atomised into tiny droplets. They are very small so they heat up very quickly and then start to burn. The expanding gases force the piston down the cylinder, thus turning the crankshaft. During this stroke work is put into engine.

EXHAUST. When the piston approaches the bottom of the cylinder, known as Bottom Dead Centre (BDC), the exhaust valve starts to open and the hot gases are expelled from the cylinder.

TWO - STROKE CYCLE

This operation cycle is so called because it takes two strokes of the piston, or one revolution of the crankshaft, to complete the process needed to produce power. In this cycle, each event is accomplished in a very short time. Moreover, the engine requires some special arrangements. First, the fresh air is forced in under pressure. The incoming air is used to clean out, or scavenge, the exhaust gases and then fill the space with fresh air charge. Instead of valves, there are special holes, called ports, which are opened and closed by piston sides as it moves up and down.

So the piston is at the top of its stroke after fuel injection and combustion have taken place. The piston is then forced down on its working stroke with the valves in the cylinder head opening the exhaust port. The burnt gases then begin to be expelled and the piston continues down until it opens the inlet or scavenge port. Next pressurised air enters and drives out the remaining burnt gases. The piston closes these ports as it returns. The air is then compressed as the piston moves to the top of its stroke. This is the explanation for the name “two stroke”, with a downward power stroke and an upward compression stroke. A two-cycle engine, therefore, has two power strokes for every one of a four-cycle engine.

FOUR - STROKE DIESEL ENGINE

The engine is made up of a piston that moves up and down in a cylinder liner which is sealed from the top by a cylinder head. The fuel injector, through which fuel enters, is located in the cylinder head. The inlet and exhaust valves are also housed in the cylinder head and held shut by springs. The piston is joined to the connecting rod by a piston pin. The bottom end, or big end, of the connecting rod is joined to the crankpin, which forms part of the crankshaft.

The crankshaft is arranged to drive the camshaft through gears. The camshaft either directly or through pushrods operates rocker arms which open the valves at the correct point in the cycle.

The crankshaft is surrounded by the crankcase and the engine block that supports the cylinders and houses the crankshaft bearings. The cylinder and the cylinder head are arranged with water-cooling passages around them.

The four-stroke has certain advantages over a two-stroke, which include higher piston speeds, wider variations in speed and load, cooler pistons, no fuel lost through exhaust and lower fuel consumption. It also consumes less lubricating oil.

TWO - STROKE ENGINE

This engine is made up of a piston that is solidly connected to a piston rod. The piston rod is attached to a crosshead bearing at the other end. The top end of the connecting rod is also joined to the crosshead bearing. Ports are arranged in the cylinder liner for air inlet and for a valve in the cylinder head that enables the release of exhaust gases. The crankshaft is supported within the engine bedplate by the main bearings. A-frames are mounted on the bedplate and house guides in which the crosshead travels up and down.

Some of the engine power is used to drive a blower that forces the air charge into the cylinder under pressure. Additionally, because of a much shorter period the intake ports are open (as compared to a four-stroke cycle), a smaller amount of air is admitted.

The main difference between the two engines is the power developed. The two-stroke engine, theoretically, develops twice as much power as the four-stroke one. Inefficient scavenging, however, reduces the power advantage.

DIESEL ENGINE TYPES

A diesel engine operates with a fixed sequence of events which are achieved either during four or two strokes. A stroke is defined as the distance the piston travels between its top and bottom points.

Various engine designs can also reflect the way the piston acts. According to this, diesel engines may be classified as single acting, when one side of the piston and one end of the cylinder are used to develop power, and double acting, if both piston sides and both cylinder ends are used to produce power.

Considering the way the piston is attached to the upper end of the connecting rod we can distinguish two types: a trunk-piston engine (if the piston is directly connected with it) and the crosshead engines (if indirectly connected).

Diesel engines usually have three general speeds ranges, in which they are classified: low -speed diesels - 50 - 300 rpm, medium-speed diesels - 300 - 1000 rpm, and high -speed diesels - above 1000 rpm.

According to their drive, engines may be classified as direct-coupled engines, i.e. coupled directly to the propeller shaft (also called direct drive engines) and geared engines, i.e. coupled to a reduction gear mechanism (indirect drive engines). If engines can rotate in both clockwise and anticlockwise direction, they are known as reversible engines. When they cannot run in the opposite direction, they are called non-reversible.

DIESEL ENGINE PARAMETERS

In general, diesel engine parameters are defined in terms of the following characteristics:

- **Cylinder Bore.** This is measured in millimetres (mm) or in centimetres (cm) and is used to identify the inner (inside) diameter of the cylinder.
- **Stroke or the Length of the Stroke.** This is the distance a piston travels between top and bottom dead centers (TDC and BDC); measured in mm or in cm.
- **Piston Speed.** This is measured in revolutions per minute (**rpm**). It is the speed at which the crankshaft rotates. Since the piston is connected to the shaft, the rpm, along with the length of the stroke, determine the piston speed.
- **Mean Piston Speed.** This describes the average speed of the piston during one rotation of the crankshaft. It is measured in metres per second (m/s).
- **Mean Effective Pressure (MEP).** This is the average pressure exerted on the piston during each power stroke and is usually defined as force per area unit, e.g. in pounds per square inch (psi), or bars.

- **Horsepower (hp).** This is the power developed within a cylinder and can be calculated by measuring the MEP and the engine speed. It is normally defined in kilowatts.
- **Stroke-to-Bore Ratio.** This is the length of the stroke divided by the cylinder bore. If the result is about 1, we can classify the engine as a square one.
- **Maximum Continuous Rating (MCR).** This is the designed maximum power which a diesel engine is capable of delivering continuously, at nominal maximum speed, in the period between two consecutive overhauls.
- **Fuel Consumption.** This depends upon the power developed by the engine. The rate of fuel consumption is the amount of fuel used in a unit of time, e.g. tons/day.

DUAL FUEL SYSTEMS

There are engines that can run on either natural gas or fuel oil. They are called dual-fuel engines. For example, the Wartsila 32DF and Wartsila 50DF, both four-stroke engines, are designed to be fuel-flexible. Changing from one fuel to the other can be done under all operating conditions.

They operate on the lean-burn principle, where the mixture of air and gas in the cylinder has more air than is needed for complete combustion. Lean combustion reduces peak temperatures and consequently NO_x emissions. It is started by injecting a small amount of LFO (called pilot fuel) into the cylinder. The pilot fuel is ignited in a conventional diesel process. Each cylinder is individually controlled to ensure the desired air-fuel ratio together with the necessary amount and timing of pilot fuel injection.

The fuel system of the Wartsila 50DF has been divided into two subsystems: one for gas and the other one for diesel oil. The engine is started in diesel mode using both main diesel and pilot fuel. At about 300 rpm, the main diesel injection is disabled and the engine is transferred to gas mode.

Natural gas is supplied to the engine through the valve station. The gas is first filtered to ensure a clean supply. The gas pressure is controlled by a valve located near the

engine driving end. Its pressure depends on the engine load. The system includes the necessary shut-off and venting valves to ensure safe and trouble-free gas supply.

On the engine, the gas is supplied through large common-rail pipes running along the engine. Each cylinder then has an individual feed pipe to the gas admission valve on the cylinder head.

The fuel oil supply system has been divided into two parts: one for the pilot fuel and the other for backup fuel. The pilot fuel is elevated to the required pressure by a pump unit. This includes filters, the pressure regulator and an engine-driven piston-type pump. The high-pressure pilot fuel is then distributed through a common-rail pipe to the injection valves at each cylinder. The pilot fuel is injected at approximately 900-bar pressure.

The back-up fuel is fed to a normal camshaft-driven injection pump. From the injection pump, the high-pressure fuel goes to a spring-loaded injection valve of standard diesel design.

JERK PUMP INJECTION

In the jerk pump injection system, there is a separate injector pump for each cylinder. The injector pump is usually operated once every cycle by a cam on a camshaft. The barrel (cylinder) and plunger (a spring-loaded ram) of the injector pump are designed to suit the engine fuel requirements. Ports (holes) in the barrel and slots in the plunger serve to regulate the fuel delivery. The pump elements (cylinder and plunger) are built into the injection pump body. As the cam rotates, it operates a spring-loaded ram (the plunger), which moves up and down in a barrel. As the plunger moves up the barrel, the pressure of the fuel in the barrel above the plunger rises very quickly. The high-pressure fuel then opens the fuel valve (the injector) and fuel is sprayed into the engine cylinder.

Different engine manufacturers use different methods to change fuel injection timing when an engine operates under part load conditions to achieve saving in fuel. This is called Variable Injection Timing (VIT).

For example, the Wartsila 64 engine uses a fuel pump with two plungers and two barrels with common suction and discharge. The plunger for controlling the start of

injection has a helix in the top of the plunger, while the plunger for controlling the end of injection is a conventional scroll-type fuel pump plunger. The same cam operates both plungers. Thus, the injection can be freely adjusted independently of the injected quantity. No lubricating oil is required for the pump element, since the plunger has a wear-resistant, low-friction coating.

LUBRICATING OIL SYSTEM

The lubrication system of an engine supplies lubricating oil to various moving engine parts. Its primary function is to form the oil film between moving parts, thick enough to reduce friction. Insufficient lubrication may cause sticking of piston rings, overheating of bearings and excessive engine wear.

The performance of modern diesel engines depends on the effectiveness of their lube oil systems. To be effective, such a system should successfully perform the following functions:

- it should control friction between load-bearing surfaces;
- it should reduce wear by preventing metal-to-metal contact between moving parts;
- it should limit the temperature by taking some of the heat away;
- it should reduce corrosion by coating metal surfaces;
- it should dampen mechanical vibrations;
- it should help to seal cylinder walls.

The lubricating system of a diesel engine can be divided into two subsystems: the one operating inside the engine, called the internal lubrication system, and the one that functions outside the engine, called the external system.

The internal system mainly consists of passages, piping, valves and filters, sometimes pumps. The external one includes such parts as tanks and sumps, pumps, coolers, strainers and filters.

OPERATING PROCEDURES

These are verbs which are commonly used in standard trouble-shooting orders and operating and maintenance procedures.

- activate
- adjust
- attach
- check
- change
- clean
- close
- connect
- correct
- disconnect
- dismount
- examine
- install
- lift
- locate
- loosen
- lower
- lubricate
- make sure
- mount
- open
- overhaul
- place
- position
- raise
- reduce
- relieve
- remove
- replace
- screw
- shut off
- start
- switch off
- switch on
- take out
- tighten
- uninstall
- unscrew

FUEL OIL SUPPLY SYSTEM

The design of the supply, also called external, fuel system varies from ship to ship, but every system should provide well-cleaned fuel with the correct temperature and pressure to each engine. The arrangement of the system usually allows for the use of both diesel and heavy fuel oil.

First, the oil is pumped from the bunker tanks or normal fuel tanks. Both of them are used for the storage of fuel oil and can be called storage tanks.

The oil is pumped to an intermediate tank, also called a settling tank. Sometimes the name “collecting tank” is used. Then it is led through heaters and centrifuges for purification. Centrifugal purifiers are used for purification of both fuel and lubricating oil.

After passing through centrifuges, cleaned and pre-heated oil is transferred to the respective service tanks, also called “day” or ‘daily’ tanks. They are used in alternation. When one tank is in use, the other one is being filled. Next, an engine-driven pump (also called a booster, transfer or primary pump) forces the fuel, from the particular service tank in operation, through the viscosity regulators, filters and flow meters to the internal (injection) fuel system. This is done by means of circulating pumps.

The system must include various safety devices such as alarms and remotely operated tank outlet valves which can be closed in the event of fire.

Part 2, ENGINE ROOM LAYOUT

ENGINE ROOM PLATFORM ONE

Starting Air Receiver

This is a kind of a storage vessel that resembles a bottle, containing compressed air for starting the main propulsion unit. Air is pressurised in air compressors and supplied to the air receivers. There are always two separate air receivers in the starting air system.

ME Emergency Control Station

This is designed to operate the main engine (ME) in the event of automatic control system failure. Such a station can be located either on the machine itself or close to it. It is normally used in case of emergency.

Main Engine

The name Main Engine (ME) currently refers to the prime source of power converting heat energy into mechanical power needed mainly to rotate the propeller and, consequently, to drive a ship.

Engine Room Platforms

Around the main engine, there are several decks in the form of galleries to provide access to all machinery. The decks are made of steel plates and gratings and must allow clear observation of the spaces below and above.

Main Engine

This is a slow-speed, two-stroke, reversible engine manufactured by MAN B&W (under licence from H. Cegielski). The type is the 7S 60 MC-C series. The engine drives a fixed pitch propeller.

Vacuum Priming Unit for Ballast Pumps

This device is designed to remove the air from the working space of the ballast pumps in order to start their operation.

Starting Air Compressor

Compressed air has many uses on board a ship, ranging from starting diesel engines to cleaning machinery during maintenance procedures. The air can be compressed in a multi-stage process.

Starting Air Compressors

The majority of engine room machinery must be doubled for emergency backup.

ENGINE ROOM PLATFORM TWO

Fuel Oil Separator (guess task)

Auxiliary Generating Sets

Auxiliary gen sets are designed for continuous operation. The gen sets consist of auxiliary engines connected to generators in order to produce electricity on board. All the sets are equipped with a complete instrument panel, alarm sensors and shutdown equipment.

Fuel Oil Separator

Marine residual fuel oil is the residue remaining after all lighter fractions have been extracted from the crude oil during various processes at the oil refinery. It often contains catalyst fines which are extremely hard and erosive and can cause serious damage to the engine. The normal way of removing catalyst fines and other contaminants from marine diesel oil is by centrifugal separation.

Auxiliary Gen Sets (guess task)

Auxiliary Engines

Auxiliary engines, together with alternators, are mounted on a common bedplate and are installed on shock absorbers.

Central Fresh Water Coolers

The fresh water is reused continuously for engine cooling. The water circulates throughout the engine cooling spaces. The water is then led to fresh water coolers, where its temperature is reduced by sea (raw) water.

Hydrophore Unit

The hydrophore provides the non-stop supply of fresh water under required pressure into the sanitary water system.

Working Air Compressor and Fresh Water Cooling Pumps

This device is designed to provide air at a specific pressure to the automatic control system of the engine room.

Fresh Water Coolers (guess task)

Fresh Water Generator

This is also called an evaporator, as it changes sea water into vapour. First, sea water enters a cylinder containing coils. Next, steam passing through the coils causes the water to evaporate.

The water vapour is then condensed and used for various applications.

Fresh Water Cooling Pumps (guess task)

Gen Sets Electric Generators

Exhaust Gas Collector

The Exhaust Gas Collector (or receiver) is an integral part of the main engine. Gaseous products of the combustion process are collected in it to avoid pressure differences resulting from rapid exhaust gas emission. Exhaust gases are the major source of waste heat and noxious emissions.

Spare Cylinder Cover

The cylinder cover, or cylinder head, is usually made of solid steel with bored passages for cooling water, and the central bore for the exhaust valve, and also separate bores for fuel valves, the safety valve and the starting valve.

ME Cylinder Heads (guess task)

Exhaust Gas Boiler

Exhaust gas boilers on diesel-propelled ships generate steam through the process of water evaporation, exploiting the high temperature of exhaust gases. The boiler

works as the heat exchanger and raises steam temperature and pressure in its own drum.

Oil-fired Boiler Burner

A boiler is a closed vessel for boiling water to create steam. The oil-fired boiler is of the water tube (or "water in tube") type. In a water tube boiler, water circulates through a series of drums and small-diameter tubes, while hot gases pass around them. The function of the burner is to provide the fuel burning process inside the boiler automatically.

Fuel Block and Hotwell

The hotwell is a steel chamber which stores the condensed exhaust steam from the oil-fired boiler in an open feed system. The fuel block houses fuel pumps, filters, heaters, viscometers and other equipment needed for proper fuel preparation.

Exhaust Gas Boiler Water Circulation Pumps

The purposes of these pumps is to ensure the continuous flow of water through the exhaust gas boiler. They are of the centrifugal type.

Centrifugal pumps make use of kinetic energy to move the liquid by means of an impeller and a circular pump casing.

The impeller produces the liquid velocity and the casing forces it to discharge from the pump converting velocity to pressure.

Oil-fired Boiler

Most oil-fired boilers are now supplied as a set including the oil burner, fuel pump, feed pumps and automatic controls for the whole unit.

Cylinder Covers (guess task)

Main Switchboard

The Main Switchboard (MSB) ensures the non-stop distribution of electricity all over the ship.

Telegraph

The telegraph is a hand-operated device located in the control stations of both the engine room and the bridge. It transmits changes in engine speed while

manoeuvring. Before START, the telegraph handle should be moved to the position corresponding to the order from the bridge.

Engine Control Room

Workstations with many computers, monitors and keyboards are installed to enable the optimal control of the ship's systems and machinery. Correct maintenance of the condition of machinery enables maintenance schedules to be planned in order to avoid breakdowns.

Cylinder Head with Exhaust Valve

Combustion gases leave the engine through exhaust valves. Each cylinder is equipped with an exhaust valve which is mounted in a central bore in the cylinder cover. The valve housing is held in place by means of four studs and nuts to form a gas-tight seal.

THE ENGINE ROOM FLOOR

Stern Tube Sealing

This special type of sealing is arranged to prevent the entry of sea water and also the loss of lubricating oil from the stern bearing.

Shaft Line

The main propulsion shaft line consists of shafting sections connected by means of bolted flange couplings and supported by bearings that keep the shafting in the proper alignment. When in service, the shafting can be checked for static and dynamic alignment. The static check can be done by clocks or strain gauges and the dynamic alignment by measuring the amplitude of axial vibrations at various speeds when the ship is underway.

Intermediate Shaft

The intermediate shaft has flanges at each end and is supported by bearings. There may be one or more sections of intermediate shafting between the crankshaft of the main engine and the tailshaft (the propeller shaft), depending upon the machinery space location.

Shaft Line Connecting Flanges

Flanges are usually circular metal plates with a ring of bolt holes to couple them together.

Shaft Line Main Bearing

This bearing supports the intermediate shafting between the tailshaft and the main engine. It is designed to provide lubrication between the bearing itself and the rotating shaft.

Main Lubricating Oil Pumps

They deliver lubricating oil to the main engine in order to minimise friction and wear between moving engine parts. Pumps can raise liquids from a lower level to a higher one.

Shaft Line

All the shafting sections are manufactured from solid steel with integral flanged couplings. The shafting sections are joined by fitted bolts made of solid steel.

Bilge System Pump and Oily Water Separator

The bilge pump is usually of the piston type. A piston is used to build up pressure in a cylindrical chamber to force the liquid through the pump. Ships must have an emergency bilge pump that can function properly in case of bilge space flooding.

Lubricating Oil Filters

Lubricating oil filters, always working in pairs, are used to remove the smallest particles of dirt from oil before it enters the moving elements of the machinery.

Bilge Piston Pump

The pump belongs to the group of displacement pumps inside which the liquid is mechanically compressed by decreasing the volume of its chamber. It is designed with a metal piston to achieve the liquid compression.

Oily Water Separator

Oily water separators are used to prevent discharging oil when ships pump out bilges outboard. Such units make use of the gravity force system together with some multi-stage filters.

Oily Water Separator

There is an automatically controlled valve which releases the separated oil to a drain tank. The nearly oil-free water leaves the separator. The water that still contains oil goes through the separating process again.

Sewage Treatment Plant

International legislation bans the discharge of untreated sewage outboard. Marine treatment plants can employ either chemical or biological methods. The biological system breaks down the sewage completely so the remaining substance can be freely discharged into any waters.

Sewage Treatment Plant, Oily Water Separator, Piston Bilge Pump (guess task)

Sea Chest (Kingston Filter)

A sea chest is a special built-in sea water filter located below the waterline. The sea water that enters the ship is filtered inside the sea chest. The sea water is used for cooling and firefighting purposes.

Ballast Pumps

Ballast pumps are usually electric-driven units mounted vertically. The liquid enters the centre of the rotating impeller enclosed within a casing and flows radially out through the vanes. The ballast system is to ensure that water can be drawn from any tank or the sea and discharged to any tank or the sea as required for trimming and heeling the ship.

A-frame of ME and Torsional Vibration Damper

The A-frame is usually mounted on the bedplate of a two-stroke engine. It is a part of an engine construction in the shape of the letter A. A torsional vibration damper is designed to absorb, or dampen, torsional vibrations that are always **present** during the engine operation.

Bilge/Fire Pumps and Sea Water Cooling Pumps

All the pumps produce a flow of fluid under pressure. The piping systems are often interconnected and most pumps installed appear in pairs. Pump failures can prove disastrous in case of emergency on board.

ME Lubricating Oil Cooler

Coolers on board ships are heat exchangers where a hot liquid is cooled by sea water. However, fresh water cooling is sometimes used to avoid corrosion problems. The plate-type cooler consists of a number of sealed titanium plates screwed together into a frame. The plates are extensively corrugated to improve the heat transfer process.

Ballast Pumps and Sea Chest (guess)

Main engine lub oil cooler (guess)

Lub Oil Filters and Separators

The centrifugal separator is used to separate oil and water, or a liquid and solids in the contaminated oil. The separation of impurities and water from oil is essential for good lubrication. The removal of contaminants from lub oil reduces engine wear and possible machinery breakdowns.

Lub Oil Duplex Filter

This is an assembly unit of two parallel filters with special valving design for the selection of full flow through either of the filters. Generally used in lubricating oil lines to allow for changeover without the stopping the flow.

Lubricating Oil Separator

The use of a centrifuge speeds up the separation process. A centrifuge separating two liquids is known as a "purifier". A centrifuge separating small amounts of water and impurities is known as "clarifier". As the device rotates at high speeds, it should be perfectly balanced and all its parts handled with special care.

ENGINE ROOM, PART 3

Interview with Peter Lund from Wärtsilä Land and Sea Academy

Part 1

Well, I came originally from Australia. I'm an Australian citizen, a permanent resident in Finland. I've been here for 7 years, living in Turku, and I work for Wärtsilä, or "Wartsila" as they say in English because we say as it...as it is written. And I'm involved in at the present time in the training business. We have a global training infrastructure called Wärtsilä Land and Sea Academy, and I work as a solutions development manager, and have done for the last two years.

The Land and Sea Academy is a Wärtsilä, it's part of the service business. Wärtsilä is mainly divided into 3 sectors or business sectors which is Marine, Power and Service business, and Training is part of the Service business.

The training...we, we have full scope or have developed into providing a full scope of training needs for our customers, which not only includes our product. It started off with the training switch in mainly providing product training for internally and for customers and then, because we have this global infrastructure we have a global advantage over individual training institutions which are resident in one country.

Part 2

In the following audio clip Peter Lund talks first about his least favourite tasks as an engineer and then about what he finds is most important.

I guess it would probably be finding the tools and equipment that had dropped down into the bilge. I'd say that would probably be the worst.

Well, I think these days the regulations and the people management, if you can do this properly, and allocation of resources is probably the major thing. You can always, it's good to have technical knowledge on everything but sometimes it's very difficult because machinery changes so much, electronics change so much and you can always, if you need to know, you can always find this information, but it's very hard to, on the spur of the moment, find managing skills and find knowledge to satisfy

regulatory bodies like survey authorities and everything so I guess the administration is probably the Chief Engineer's biggest task now.

I would say, the major thing, it's very good, the life at sea. As far as growing the mind, there's nothing like travel for broadening the mind and I still believe this but in this day and age I'd say the biggest advice I could have to any young person is to be aware.

Fuel Bacteria Results in Grounding

Narrative

When due for her annual overhaul, one of the Skye based inter-island ferries *Loch Striven* was to be replaced by another vessel from the Clyde. The two vessels met at Kyle of Lochalsh, where the crews changed over. Apart from the motorman who remained onboard, the delivery crew was now manning the southbound ferry. The voyage south was to be made in stages, dependent on weather conditions, with overnight stops as required. The chargehand in command was experienced and qualified, and had regularly undertaken delivery voyages in this area.

The first leg of the voyage passed without any problem and she spent the night in Oban. The forecast on the following day was for wind strength 4 to 6 increasing to force 7 in the evening. It was decided she could reach Campbeltown by nightfall and before the weather deteriorated. *Loch Striven* sailed from Oban at 0730 for the Clyde. By the time she cleared the shelter of the Kintyre peninsula the wind had increased to force 6. At 1755, the forward main engine stopped due to choked fuel filters. These were cleaned, the engine restarted, and the vessel resumed her voyage. As she rounded the Mull of Kintyre and turned north, the wind increased further. While lining up the leading lights for entry into Campbeltown Loch at 2110, the aft main engine stopped. Soon afterwards she grounded on Macringan's Point under the influence of beam seas and the wind. Again the cause of the engine stoppage was found to be choked fuel filters. These were cleaned and the engine restarted. With both main engines now operating and the hull only slightly damaged, the vessel came off the ground and made her way safely into harbour under her own power where she tied up at 2140.

The subsequent investigation found that the fuel tank was contaminated with bacteria, which had formed black deposits on the internal tank structure. The deposits had broken away during the rough weather and had entered the fuel systems. These built up in the fuel filters and eventually led to fuel starvation. The frequency of choked filters had been recorded while in service and the question of fuel tank cleaning was to be investigated during the overhaul.

The owners later established that the "blocked fuel filter" problem, although known to the motorman, had not been identified on the handover notes. This potential problem was not therefore known to the chargehand in command. The choking of the fuel filters, although of the duplex type allowing for quick changeover, only became known when the main engine revolutions started to drop. On the approach to Campbeltown, both motormen were in the accommodation, and by the time they became aware that there was a problem, the engine had stopped.

The Lessons

1. If fuel filters become blocked, find out why!
2. Make sure that whoever is in command is made aware of the problem, the estimated time it should take to fix, and whether it was an intermittent or regular feature.
3. If fuel starvation is a possibility, someone should be closed up in the engine room while negotiating narrow or dangerous waters.
4. When changing crews or personnel, the handover procedure **MUST** include any known defects, details of all known machinery problems, and an indication of difficulties likely to be experienced.

Footnote

If bacterial contamination of fuel and fuel tanks is suspected, take samples and have them tested as soon as possible. Cleaning fuel storage tanks and using biocides will overcome the problem and prevent a recurrence. It is a good idea to eliminate all water from fuel tanks by regularly draining them to remove condensation and any water delivered with bunkers.

Source: *Safety Digest - Lessons from Marine Accident Reports 2/2000*. Department of the Environment, Transport and the Regions, the United Kingdom. Marine Accident Investigation Branch.

A maintenance procedure

Injection Valve

The liquid fuel injection system e.g. of a diesel engine, consists of injection pump, high pressure pipe and injection valve. The injection valve injects fuel into the cylinder of the engine. It is important that the distribution of the fuel spray is as even as possible. Checking the distribution is a standard maintenance procedure of an engine.

Checking the spray distribution

The symmetrical distribution of spray can be evaluated when having the opening pressure adjusted to 50-100 bar. The needle stroke using a hand test pump is close to nominal at this low opening pressure.

ENGINE-ROOM FIRE ALONGSIDE OIL BERTH

Narrative

An oil tanker of 2,979 deadweight tonnes was using a cargo pump to discharge ballast to shore facilities. The pump was driven by an auxiliary engine located in the engine-room. A connecting rod punctured the engine entablature which caused a fire to start. As soon as the engine-room smoke detection alarm was activated the general alarm was sounded and the port emergency plan was initiated. The fire was eventually extinguished by the shore fire brigade using high expansion foam. There were no resultant injuries to personnel.

Observations

1. The connecting rod of the auxiliary engine probably became detached due to a fractured rocker arm, preventing the opening of the exhaust valve, and subsequent overloading on the respective piston.
2. The cause of the fire is deduced to have been an ignition of crankcase oil vapour by the hot white metal of the bottom end bearing.
3. Operation of the remote pull-wire arrangement failed to initiate the gang release of the fixed CO2 fire extinguishing system. A local attempt to release it was aborted when the CO2 bottle room had to be evacuated due to leakage from a joint on the pressure alarm sensor fitted to the gas manifold.

4. The minimal forward draught of the vessel prohibited an intake of water from the sea to the emergency fire pump. An alternative intake from the forepeak was unavailable because the tank was empty.
5. The auxiliary engine could not be stopped because the bridge remote stop arrangement, although installed, was not connected. The fuel tank quick-closing valves were shut but, because the fuel in the common fuel line to the multi-engine installation was of sufficient capacity, the auxiliary engine continued to run for a prolonged period.
6. The engine-room ventilation trunking had years of accumulated oil internally and thick coats of paint externally which contributed to the intensity of the fire. The engine-room had been in a dirty condition.
7. A crew member had to be rescued from his cabin after failing to respond to the sound of the general alarm. A number of false alarms had occurred in the past and he assumed that this was another one.

Comment

The management company has been advised to:

1. Clean the engine-room and engine-room ventilation trunking;
2. Ensure that auxiliary engines can be shut down by the intended remote means;
3. Pressure-test the CO2 manifold and fittings to the working pressure;
4. Conduct periodic checks to ensure the security of the CO2 pressure alarm sensor;
5. Confirm the effectiveness of the remote control wire for the CO2 release;
6. Maintain a log of fire alarm initiations in order to ascertain the frequency of false alarms;
7. Give an assurance that the emergency fire pump will always be available to maintain pressure on the fire main;
8. Issue standing orders for crew members to respond promptly to any emergency alarm.

Source: Marine Accident Investigation Branch (MAIB) - Safety Digest 03/1996